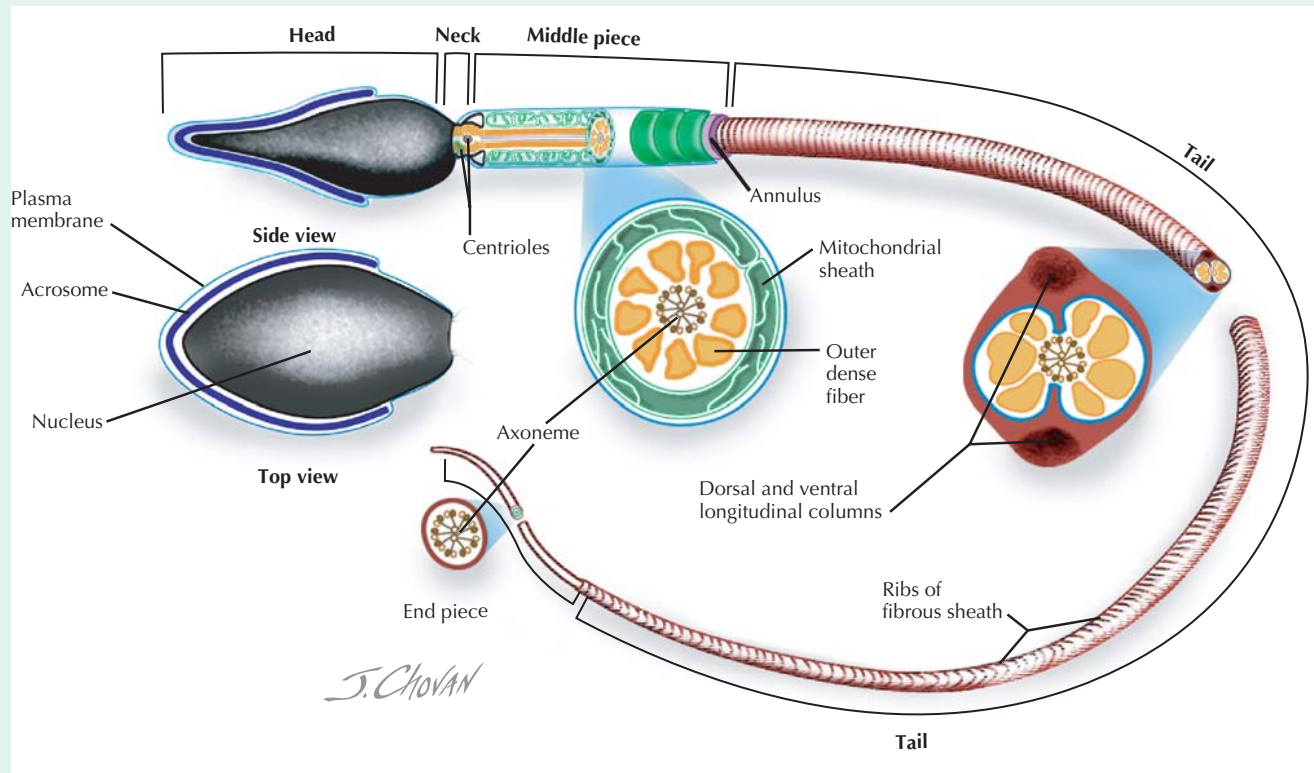




Spermatogenesis is a complex process that begins with mitotic division of spermatogonia to form primary spermatocytes, along with additional spermatogonia (unlike women, in whom there are a fixed number of primary oocytes by the time of birth, the number of germ cells in men is not fixed). Meiosis of primary spermatocytes produces secondary spermatocytes, and subsequently, haploid spermatids, which undergo further differentiation to become spermatozoa. The final product, the mature spermatozoon, includes a head piece, a mitochondria-rich middle section for energy production, and a motile tail section. The head piece has a prominent acrosome containing proteolytic enzymes necessary for penetration of an ovum.

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Mature Spermatozoon The morphology of mature spermatozoa reflect their motility and ovum-penetrating functions.

ejaculation. Sertoli cell processes form tight junctions that provide a “blood-testis” barrier, sheltering the developing spermatocytes from potential damage by blood-borne substances.

Endocrine Regulation of Testicular Function

Testicular function (both gametogenesis and steroidogenesis) is regulated by the hypothalamus and pituitary gland (Fig. 31.8). GnRH is secreted by hypothalamic nuclei into the hypophyseal portal circulation (as in females, GnRH release is pulsatile in males), and subsequently stimulates the release of LH and FSH by anterior pituitary gonadotrophs. LH stimulates the first step in testosterone synthesis (conversion of

cholesterol to Δ^5 -pregnenolone by CYP11A10, whereas FSH acts on Sertoli cells, stimulating the synthesis of **androgen-binding protein**, which subsequently binds testosterone, promoting spermatogenesis in the seminiferous tubules (see Fig. 31.7). Negative feedback in the hypothalamic-pituitary-testis axis is effected by inhibition of GnRH and LH release by testosterone and by inhibition of FSH secretion via inhibin produced by Sertoli cells.

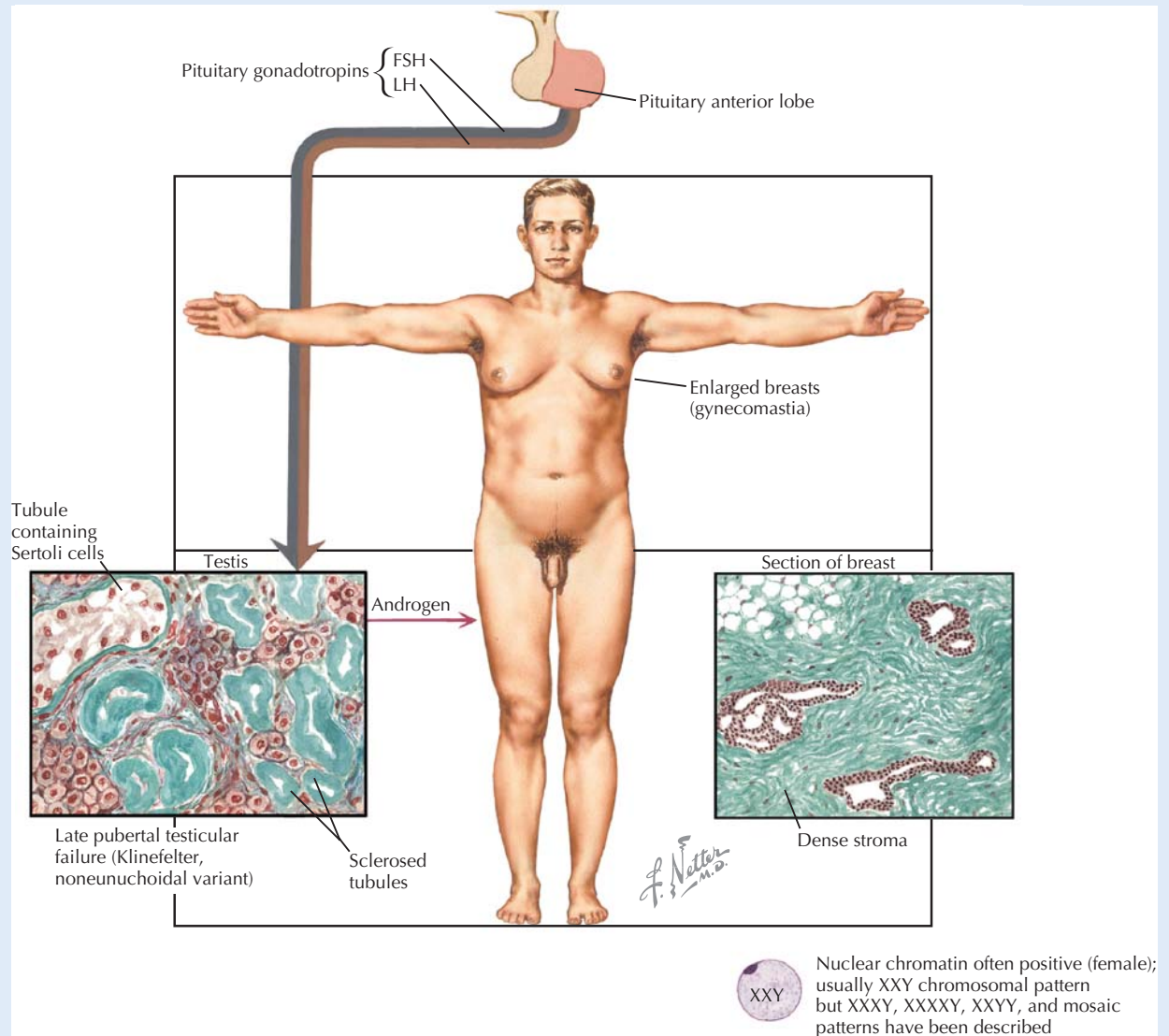
Nonreproductive Actions of Testosterone

In addition to the nonreproductive actions of testosterone at puberty discussed, increased muscle mass, development of the male pattern of hair distribution and baldness, and deep-

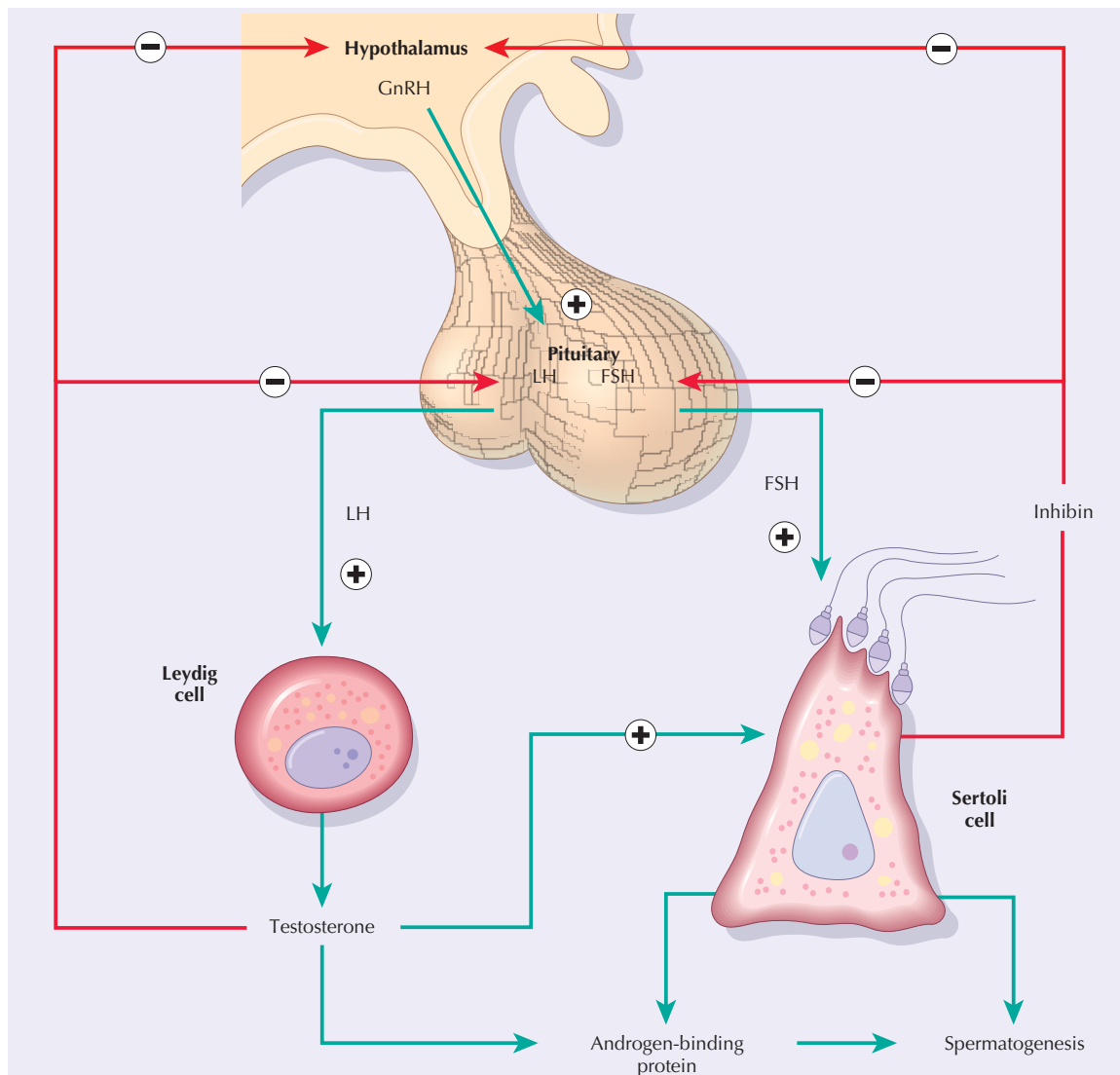
CLINICAL CORRELATE Klinefelter's Syndrome

Male infertility has a large number of causes, including blockage of the vas deferens due to infection or injury; low sperm count (oligospermia) due to infections such as mumps and some sexually transmitted diseases; lack of sperm production as a result of chronic disease, hormonal problems, or injury to the testes; impotence; and exposure to toxins. In some cases, infertility may have

a genetic basis. For example, Klinefelter's syndrome is characterized by the genotype 47, XXY. Patients with this syndrome are phenotypic males but have some identifiable physical characteristics including gynecomastia (breast enlargement) and are nearly always infertile (see illustration). They are hypogonadal, meaning that gonadal hormone levels are low, and as a result, gonadotropin levels are elevated and testicular size is reduced. Klinefelter's syndrome affects one to two men per 1000.



Testicular Failure in Klinefelter's Syndrome The presence of two X chromosomes (47, XXY genotype) results in seminiferous tubular dysgenesis and infertility, as well as primary hypogonadism (low testicular hormone levels, and as a result, high gonadotropin levels).



J. Perkins
MS, MFA

Figure 31.8 Control of Testicular Function GnRH secreted by the hypothalamus stimulates LH and FSH secretion by the anterior pituitary. LH stimulates testosterone synthesis by the Leydig cells of the testes, whereas testosterone and FSH are required for spermatogenesis. FSH induces production of androgen-binding protein (ABG); androgen is concentrated in the tubules by binding to ABG, promoting spermatogenesis. Sertoli cells also produce inhibin, which, along with testosterone, exerts negative feedback effects on the axis (inhibin specifically inhibits FSH secretion). FSH, follicle-stimulating hormone; GnRH, gonadotropin-releasing hormone; LH, luteinizing hormone.



Sex hormone-binding globulin (SHBG) and androgen-binding protein (ABG) are identical proteins, produced in the liver and testes, respectively. SHBG circulates in the bloodstream and is the major carrier for testosterone and estradiol in the blood (only a small fraction of sex hormones circulates as free hormone). ABG is produced by the testes and serves to concentrate testosterone within the seminiferous tubules, where androgens promote spermatogenesis.

ening of the voice are also actions of testosterone. Some of the actions of testosterone are dependent on its conversion to **dihydrotestosterone** (genital differentiation, prostate development and growth, male hair distribution, and baldness, for example), while others are direct effects of testosterone.

Review Questions

CHAPTER 26: GENERAL PRINCIPLES OF ENDOCRINOLOGY AND PITUITARY AND HYPOTHALAMIC HORMONES

1. Which of the following hormones initiates its actions at target tissues by binding to nuclear receptors?

- A. ACTH
- B. Vasopressin
- C. Epinephrine
- D. IGF-1
- E. Thyroid hormone

2. Hormonal actions are regulated by:

- A. affinity of receptors for the hormone.
- B. the number of receptors for the hormone.
- C. the degree of binding of hormone to plasma proteins.
- D. the concentration of hormone in the blood.
- E. All of the above.

3. Vasopressin (ADH) secretion by the posterior pituitary is stimulated by:

- A. decreased body fluid osmolarity.
- B. decreased blood volume.
- C. atrial natriuretic peptide.
- D. ethanol.
- E. high blood pressure.

4. The hormone oxytocin:

- A. is released directly into the general circulation.
- B. is synthesized in the posterior pituitary gland.
- C. promotes uterine quiescence.
- D. has a well-established physiological role in induction and progression of labor.
- E. is a 191 amino acid peptide.

5. Growth hormone secretion:

- A. occurs in pulses, mainly during sleep.
- B. is stimulated by a specific hypothalamic neurohormone.
- C. is inhibited by somatostatin.
- D. results in synthesis of insulin-like growth factor in a variety of target tissues.
- E. All of the above.

CHAPTER 27: THYROID HORMONES

6. Thyroid hormone secretion is:

- A. inhibited by elevated plasma estrogen levels.

- B. stimulated by high plasma T_3 .
- C. inhibited by high thyroxine-binding globulin (TBG).
- D. stimulated by high plasma T_4 .
- E. stimulated by TSH.

7. In adults, hypothyroidism is associated with which of the phenotypes listed below?

- A. Hypertension, cold intolerance, diarrhea
- B. Hypertension, nonpitting edema, cold intolerance
- C. Hypotension, higher than normal temperature, diarrhea
- D. Hypotension, higher than normal temperature, nonpitting edema
- E. Hypotension, nonpitting edema, cold intolerance

8. In the plasma, thyroid hormone:

- A. is present primarily in the T_3 form.
- B. is present primarily in the T_3 form and is bound to proteins.
- C. is completely bound to thyroxine-binding globulin (TBG).
- D. is present primarily in the T_4 form and is mostly bound to proteins, including TBG.
- E. is present only as free T_3 and T_4 .

9. Feedback regulation of thyroid hormone is illustrated by:

- A. high circulating levels of free T_3 and T_4 , which elevate TRH.
- B. high circulating levels of free T_3 and T_4 , which elevate TSH.
- C. high circulating levels of free T_3 and T_4 , which reduce TRH.
- D. reduced TRH, which elevates TSH.
- E. reduced TSH, which elevates TRH.

CHAPTER 28: ADRENAL HORMONES

10. Stimulation of the adrenal gland by ACTH has the most direct effect on release of:

- A. epinephrine.
- B. norepinephrine.
- C. cortisol.
- D. androgens.
- E. aldosterone.

11. Which portion of the adrenal gland secretes a hormone with anti-inflammatory actions?

- A. The medulla
- B. The zona reticularis
- C. The zona fasciculata
- D. The zona glomerulosa
- E. None of the above.

12. Which of the following is a significant action of adrenal androgens in females?

- A. Stimulation of libido
- B. Hypertrophy of sebaceous glands
- C. Development of pubic hair
- D. Development of axillary hair
- E. All of the above.

13. Aldosterone secretion by the adrenal gland is stimulated by:

- A. hypokalemia.
- B. high plasma Na^+ concentration.
- C. angiotensin II.
- D. atrial natriuretic peptide.
- E. All of the above.

CHAPTER 29: THE ENDOCRINE PANCREAS

14. Insulin is necessary for:

- A. entry of glucose into tissues via the GLUT1 transporter.
- B. entry of glucose into all tissues.
- C. entry of glucose into the brain.
- D. entry of glucose into the pancreas.
- E. entry of glucose into tissues via GLUT4 transporters.

15. Regulation of insulin release by increased plasma glucose involves which of the following sequence of steps?

- A. Entry of glucose into pancreatic β -cells through GLUT1 transporters; increased ATP, which decreases K^+ efflux; depolarization of the β -cell; Ca^{2+} influx; insulin granule release
- B. Entry of glucose into pancreatic β -cells through GLUT1 transporters; hyperpolarization of the β -cell; Ca^{2+} efflux; insulin granule release
- C. Entry of glucose into pancreatic β -cells through GLUT2 transporters; increased ATP, which decreases K^+ efflux; depolarization of the β -cell; Ca^{2+} influx; insulin granule release
- D. Entry of glucose into pancreatic β -cells through GLUT2 transporters; hyperpolarization of the β -cell; Ca^{2+} efflux; insulin granule release
- E. Entry of glucose into pancreatic β -cells through GLUT4 transporters; depolarization of the β -cell; Ca^{2+} influx; insulin granule release

16. Factors that stimulate insulin secretion include:

- A. glucagon-like peptide-1 (GLP-1).
- B. amino acids.
- C. glucose insulinotropic peptide (GIP).
- D. fatty acids.
- E. All of the above.

17. Insulin is a hypoglycemic hormone because of its ability to:

- A. increase gluconeogenesis.
- B. reduce glycogen storage in muscle.
- C. increase glycogenolysis.
- D. increase GLUT1 transporters in brain tissue membranes.
- E. increase GLUT4 transporters in adipose tissue and skeletal and cardiac muscle membranes.

CHAPTER 30: CALCIUM-REGULATING HORMONES

18. Decreased plasma calcium levels result in:

- A. reduced production of 1,25-dihydroxycholecalciferol.
- B. reduced production of 25-hydroxycholecalciferol.
- C. increased bone mineralization.
- D. increased bone resorption.
- E. increased plasma calcitonin.

19. Parathyroid hormone (PTH):

- A. is released into plasma in response to increases in plasma calcium.
- B. acts at the kidneys to increase calcium reabsorption and decrease phosphate reabsorption.
- C. acts at the kidneys to decrease calcium reabsorption at the distal tubule.
- D. acts at the bone to increase bone mineralization.
- E. acts at the bone to decrease osteoclastic osteolysis.

20. Primary hyperparathyroidism is often associated with:

- A. development of renal calculi.
- B. bone pain.
- C. constipation.
- D. fatigue.
- E. All of the above.

CHAPTER 31: HORMONES OF THE REPRODUCTIVE SYSTEM

21. During fetal development, the formation of male genitalia is directly stimulated by:

- A. testosterone.
- B. dihydrotestosterone.

- C. dehydroepiandrosterone.
- D. androstenedione.
- E. estrone.

22. The pubertal growth spurt is mainly stimulated by the hormone:

- A. calcitonin.
- B. vitamin D.
- C. estradiol.
- D. testosterone.
- E. dihydrotestosterone.

23. Conversion of androgens to estradiol by the follicular granulosa cells is stimulated by the hormone:

- A. LH.
- B. FSH.
- C. oxytocin.
- D. growth hormone.
- E. prolactin.

24. During the prepubertal period, germ cells in women are arrested at which stage of gametogenesis?

- A. Oogonia
- B. Primary oocyte
- C. Secondary oocyte
- D. Ovum
- E. Zygote

25. The hormone inhibin has direct negative feedback effects on the release of:

- A. prolactin.
- B. FSH.
- C. LH.
- D. testosterone.
- E. GnRH.

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Answers

Section 1: Cell Physiology, Fluid Homeostasis, and Membrane Transport

1. **B.** Antipyrine diffuses throughout the body water and can be used to measure the volume of total body water; inulin does not enter cells and can be used to measure the extracellular fluid (ECF) volume. Using the indicator-dilution formula, the amount of indicator injected divided by the plasma concentration will result in the volume of the compartment. By this method, the total water (indicated by antipyrine) is 36 L, and the ECF (indicated by inulin) is 12 L; the intracellular fluid (ICF) is 24 L. The ECF (12 L) is one-fourth plasma (3 L) and three-fourths interstitial fluid (ISF) (8 L).
2. **D.** The Starling forces favoring movement out of the capillaries are the capillary hydrostatic pressure (HP_c) and interstitial oncotic pressure (π_i), while the forces favoring movement into the capillaries are the interstitial hydrostatic pressure (HP_i) and capillary oncotic pressure (π_c). Thus, $(30 + 8) - (3 + 28) = 7$ mm Hg out of the capillary.
3. **D.** Water will distribute through all compartments, and thus, both ECF and ICF volumes will increase. With the additional volume, the osmolality of the compartments will decrease.
4. **C.** Voltage-gated channels are highly selective, and gating is dependent on membrane voltage.
5. **D.** The basolateral Na^+/K^+ ATPase, or "sodium pump," uses primary active transport to pump three sodium ions out of cells in exchange for two K^+ ions. The low intracellular Na^+ concentration facilitates apical sodium entry down its concentration gradient through several different secondary active transport symporters (e.g., Na^+ -amino acid, Na^+ -glucose) and antiporters (e.g., Na^+/Ca^{2+} , Na^+/H^+).
6. **B.** Ouabain (digoxin) is a cardiac glycoside that irreversibly blocks Na^+/K^+ ATPase, allowing equilibration of sodium and potassium across the cell membrane. This stops sodium-dependent transport processes and depolarizes the membrane potential.

Section 2: The Nervous System and Muscle

1. **A.** Reduced permeability of sodium ion will result in hyperpolarization of the cell. Ordinarily, sodium leakage into the cell contributes to the actual resting membrane potential, making it less negative than would be predicted on the basis of potassium concentration and permeability. Reduction in potassium ion permeability, on the other hand, would cause depolarization of the cell, as would influx of calcium ion or increased extracellular concentration of sodium or potassium ion.
2. **C.** Neuronal action potentials result when the threshold potential is reached and voltage-gated sodium channels open. Although action potentials are evoked in some excitable cells by opening of Ca^{2+} channels, opening of voltage-gated Na^+ channels is responsible for the phase 0 upstroke of the neuronal action potential.
3. **D.** Myelination of an axon results in decreased capacitance and increased membrane resistance. As a result, current travels through the interior of the axon but not through the membrane. Myelination produces a large increase in conduction velocity.
4. **A.** When axonal depolarization reaches the synaptic bouton, voltage-gated Ca^{2+} channels are opened, resulting in influx of Ca^{2+} and vesicular release of neurotransmitter into the synaptic cleft.
5. **D.** A single, large Na^+ and K^+ channel conducts the depolarizing charge during an endplate potential while the neuronal action potential involves multiple ion channels and depolarization is

mainly produced by Na^+ influx. Depolarization is rapid in both endplate and action potentials. Depolarization to +40 mV, causation by voltage-gated channels, and repolarization associated with increased K^+ conductance are all characteristics of neuronal action potentials but not endplate potentials.

6. **E.** The neural tissue of the spinal cord is covered by three membranes (meninges). These are the inner pia mater, the middle arachnoid membrane, and the outer dura mater.
7. **B.** CSF is produced by the choroid plexus and is reabsorbed at the arachnoid granulations into the venous system, as well as into the capillaries of the CNS and pia mater.
8. **E.** Association pathways functionally and anatomically linking the two cerebral hemispheres include the corpus callosum and anterior, posterior, and hippocampal commissures.
9. **C.** While the pons is also involved in regulation of breathing, the medulla is the major site of autonomic regulation, including regulation and integration of cardiovascular and respiratory functions. The thalamus and hypothalamus are not part of the brainstem.
10. **A.** Meissner's corpuscles, located in the dermal papillae, especially in fingertips, palms, soles, lips, face, tongue, and genital skin (non-hairy skin), are rapidly adapting receptors with small receptive fields that allow point discrimination and detection of low-frequency stimuli such as flutter. Pacinian corpuscles respond to rapid changes in pressure and vibration, Merkel's disks respond to pressure and touch, especially indentation of the skin; hair follicle receptors respond to movement across the skin surface; and Ruffini's corpuscles respond to stretch of skin and joints.
11. **B.** Photoreceptors respond to light reception through a sequence of events that ultimately results in reduced Na^+ permeability and hyperpolarization of their cell membranes, which inhibits neurotransmitter release by the photoreceptors.
12. **E.** The three ducts of the membranous labyrinth of the cochlea of the inner ear are the scala vestibuli, scala tympani, and scala media. The scala media contains endolymph, which resembles intracellular fluid in its composition. The other two ducts contain perilymph, which resembles extracellular fluid in its composition.
13. **E.** The pattern of pressure changes between the three perpendicularly oriented semicircular canals during angular acceleration of the head is dependent on the direction of head movement. The bending of cilia as a result of these pressure changes produces hyperpolarization or depolarization and changes in neurotransmitter release by hair cells. As a result, the firing rate of afferent nerve fibers ending on the hair cells is altered. These signals are carried through cranial nerve VIII to the vestibular nuclei of the pons.
14. **A.** When the patellar tendon of the knee is tapped, muscle spindles in the quadriceps muscle are stretched, producing a reflex in which type Ia afferent nerves are depolarized, and conduct the signal to the spinal cord. The nerves synapse with α -motor neurons, which conduct the signal back to the quadriceps, causing contraction. This stretch reflex is a monosynaptic reflex. Golgi tendon reflex is a bisynaptic reflex involved in preventing muscle damage when muscle tension is excessive. The flexor withdrawal reflex is a polysynaptic reflex that occurs in response to painful stimuli.
15. **E.** The corticospinal tract carries fibers that originate in the primary motor cortex, premotor and supplemental motor areas, and somatosensory areas posterior to the motor cortex. It is the most important pathway for fine motor activity controlled by the cortex. The other

tracts mentioned are involved in other aspects of motor function, including brainstem control of balance and posture.

16. **C.** The cerebellum coordinates and fine-tunes motion. All of the efferent signals from the cerebellar cortex are inhibitory, originating from the Purkinje cells. This inhibitory output is regulated by excitatory input from climbing fibers and mossy fibers by way of the granule cells, and inhibitory input from interneurons in the cerebellar cortex.
17. **B.** Like the cerebellum, the basal ganglia have an accessory role to the motor cortex. With input from the motor cortex and output to the motor cortex through the thalamus, the basal ganglia function to produce smooth movement and regulate posture.
18. **B.** Preganglionic nerves of the sympathetic nervous system originate in the intermediolateral and intermediomedial cell columns of the thoracolumbar spinal cord (T1 to L3). Their axons extend to the chain ganglia, where they synapse with postganglionic nerves. Preganglionic nerves of the parasympathetic nervous system have cell bodies in the brainstem.
19. **D.** Postganglionic neurons of the parasympathetic nervous system (PNS) release acetylcholine, which acts at muscarinic receptors at the neuroeffector junction. Postganglionic neurons of the sympathetic nervous system (SNS) primarily release norepinephrine and epinephrine, which act at a variety of adrenergic receptors on effector organs and tissues. Both sympathetic and parasympathetic preganglionic nerves achieve ganglionic neurotransmission by release of acetylcholine, which acts at nicotinic receptors.
20. **C.** β_2 -adrenergic receptors mediate sympathetic bronchial dilation. Constriction of vascular smooth muscle and mydriasis are produced by norepinephrine and epinephrine release by postganglionic sympathetic nerves and subsequent binding to α_1 receptors; increased contractility is mediated by β_2 receptors. Sweat gland secretion is stimulated by atypical sympathetic postganglionic nerve fibers that secrete acetylcholine.

Section 3: Cardiovascular Physiology

1. **B.** The systemic veins normally contain more than 60% of the blood volume and can act as a reservoir for blood. Blood can be mobilized from veins during hemorrhage or dehydration as a compensatory measure to help maintain blood pressure until blood volume can be restored.
2. **C.** Contraction of the left ventricle consists of powerful constriction of the chamber with shortening of the heart from base to apex, resulting in the high pressures required to pump blood through the high resistance systemic circulation.
3. **A.** The coronary circulation receives approximately 4% of the cardiac output at rest, while the other listed parts of the circulation each receive approximately 15% (brain) to 24% (liver and GI tract) of the circulation.
4. **E.** Following the plateau of the action potential, gradual inactivation of L-type Ca^{2+} channels leads to activation of K^+ channels and rapid repolarization. Opening of Na^+ channels is responsible for the phase 0 upstroke; this is accompanied by reduced conductance of the inwardly rectified K^+ current. Opening of L-type Ca^{2+} channels is mainly responsible for the plateau (phase 2); inactivation of Na^+ channels and opening of voltage sensitive K^+ channels cause the rapid repolarization to the plateau (phase 1).
5. **E.** A multiple-lead electrocardiogram can yield information on cardiac rhythm and conduction; presence, location, and extent of ischemia; orientation of the heart and chamber size; and abnormalities resulting from altered electrolyte levels or drugs. It does not yield direct information regarding cardiac output, stroke volume, ejection fraction, or other flow-related parameters.
6. **B.** Conduction velocity is a function of the slope of the phase 0 upstroke of the action potential. Velocity is slowed in the AV node, resulting in a pause between atrial and ventricular depolarization.

As a result, there is a pause between atrial and ventricular contraction, allowing atrial contraction to produce the final filling of the ventricles before they contract.

7. **B.** Mean arterial pressure is not simply the arithmetic mean of diastolic and systolic arterial pressure, because the arterial pressure curve is complex in morphology and because diastole is longer in duration than systole. As a rule of thumb, at normal heart rates, mean arterial pressure is approximately equal to the diastolic pressure plus one third the pulse pressure. In this problem, the diastolic pressure is 80 and one third of pulse pressure is 16.7; therefore, the mean arterial pressure is predicted to be approximately 97 mm Hg.
8. **A.** The highest pulse pressure is observed in the left ventricle, where normal resting pressure is approximately 120/0 mm Hg. High systolic pressure is required to pump blood through the high resistance systemic circulation, while low diastolic pressure allows filling during diastole. Within the pulmonary circuit, the highest pulse pressure occurs in the right ventricle (approximately 25/0 mm Hg at rest).
9. **E.** According to Poiseuille's law, flow is proportional to the fourth power of the radius of a tube. Thus, doubling the radius of the tube will result in an increase in flow by a factor of 16, if other variables are held constant. Doubling the pressure gradient, halving the viscosity of the fluid, or halving the length of the tube will double the flow; doubling the viscosity will reduce flow by 50%.
10. **C.** High viscosity of fluid favors laminar flow. When the hematocrit is reduced, resulting in low blood viscosity, flow may be turbulent, resulting in audible murmurs. Wide tube diameter, rapid velocity of fluid flow, high density of fluid, and pulsatility of flow all increase Reynolds number and favor development of turbulent flow.
11. **C.** Wall tension (T) is defined by Laplace's law, $T = P_r r$, where P_r is the transmural pressure and r is vessel radius. Reduction of interstitial pressure will result in increased transmural pressure, and thus, increased wall tension. Reduced radius, vascular hydrostatic pressure, or transmural pressure will be associated with lower wall tension.
12. **D.** The isovolumetric contraction period begins when ventricular contraction results in ventricular pressure rising to exceed atrial pressure, causing the atrioventricular valve to close. Thus, the beginning of the isovolumetric contraction period for the right ventricle is marked by closure of the tricuspid valve.
13. **A.** During the cardiac cycle, valves on the left side of the heart close before the corresponding valves on the right side (mitral valve closes before the tricuspid, and aortic valve closes before the pulmonic), whereas valves on the right side open before the corresponding valves on the left side (pulmonic valve opens before the aortic, and tricuspid valve opens before the mitral valve).
14. **E.** In respiratory sinus arrhythmia, which occurs primarily in infants and children, heart rate is increased during inspiration and reduced during expiration. This is caused by increased venous return during inspiration, causing stretch of low pressure atrial baroreceptors and reflexive increase in heart rate.
15. **E.** Activation of the sympathetic nervous system (SNS) will produce all of the listed effects. Effects on the heart (increased heart rate and contractility) involve binding of adrenergic neurotransmitter to β_1 receptors; constriction of arteries and veins is produced by binding of the adrenergic transmitter to α_1 receptors.
16. **B.** The force-velocity relationship for cardiac muscle reveals an inverse relationship between velocity of contraction and afterload (force of contraction). With increased contractility, the maximum velocity of contraction (V_m , which occurs at zero afterload), is increased, as is the maximum force of contraction. Maximum force of contraction occurs during isometric contraction, at the x-intercept of the force-velocity relationship, where velocity is zero.

17. **B.** The vascular function curve represents the relationship between central venous pressure and cardiac output when central venous pressure is the dependent variable; the cardiac function curve represents the relationship when cardiac output is the dependent variable. The intersection of the two curves is at the normal, resting cardiac output and central venous pressure at equilibrium (approximately 5 L/min cardiac output and central venous pressure (CVP) of approximately 2 mm Hg).
18. **C.** Nitric oxide release by endothelial cells produces vasodilation of the vessel by relaxing underlying smooth muscle. This effect is mediated by the second messenger cGMP, which reduces free intracellular Ca^{2+} , producing the smooth muscle relaxation. Dilation of arterial vessels results in higher capillary hydrostatic pressure downstream. Nitric oxide also inhibits adhesion of platelets to the vascular wall.
19. **A.** In many tissues and organs, if blood flow is increased due to higher perfusion pressure, the expected elevation in flow will be followed by a return in blood flow toward the basal rate. According to the myogenic hypothesis, this autoregulation involves smooth muscle constriction in response to elevated transmural pressure (in other words, in response to stretch).
20. **D.** β_2 receptor binding produces vasodilation, whereas α_1 and α_2 receptor binding are associated with vasoconstriction. β_1 receptors are found in the heart, where the main effects mediated by these receptors are increased heart rate, contractility, and conduction velocity.
21. **A.** Arterial baroreceptors respond to high arterial pressure (and thus, stretch) by sending afferent nerve impulses to the central cardiovascular center, resulting in reduced sympathetic efferent activity and increased parasympathetic activity. In addition to high pressures, the baroreceptors also respond to pulse pressure.
22. **C.** Left coronary artery flow is highest during early diastole. Flow is low during systole, due to compression of myocardial vessels by the contracting myocardium. As the heart relaxes, this compression is released, and this, combined with the effects of vasodilator metabolites which build up in the myocardium during the low flow of systole, results in a large increase in left coronary artery blood flow in early diastole.

Section 4: Respiratory Physiology

1. **A.** A rise in pulmonary artery pressure produces passive distension of vessels in the pulmonary microcirculation and opening of some vessels that were previously collapsed (recruitment).
2. **D.** Spirometry measures changes in lung volume (tidal volume, expiratory reserve volume, inspiratory reserve volume, vital capacity, inspiratory capacity), but cannot measure total lung capacity, residual volume, or functional residual capacity. To determine these three values, one of them must be measured indirectly, for example by nitrogen washout, helium dilution, or body plethysmography.
3. **C.** In the standing position, both ventilation and perfusion of the lung are greatest in the bottom portion and poorest in the upper portion of the organ. However, the vertical gradient for perfusion is much greater than the gradient for ventilation. Therefore, the ventilation-to-perfusion ratio is highest toward the top of the lung. The ratio approaches infinity in areas of dead space and zero in areas of shunt.
4. **B.** Diffusion of a gas through a membrane is a passive process that follows Fick's law. It is directly related to the partial pressure gradient, directly related to surface area, directly related to the diffusion constant of the gas, and inversely related to membrane thickness.
5. **C.** In the middle portion of the lung, zone 2, alveolar pressure falls between pulmonary arterial and venous pressures, and the ventilation and perfusion are approximately balanced, resulting in a ratio of approximately 1.
6. **B.** Functional residual capacity (FRC) is lung volume after expiration in normal, quiet breathing. At this point, mechanical forces are in balance, with outward elastic recoil pressure of the chest wall balancing the inward elastic recoil pressure of the lung.
7. **C.** In the respiratory system as a whole, the greatest resistance to flow occurs in the medium-sized airways (fourth to eighth generation). Proceeding down the airways, the diameter of airways decreases while the number of tubes increases rapidly. Taking into consideration both factors, the resistance is greatest in the medium-sized bronchi (in aggregate).
8. **A.** With progressively greater effort, peak air flow is increased during expiration, but along the downward slope of the expiratory flow-volume curves, airflow is effort-independent.
9. **B.** Severe COPD is characterized by emphysema, with increased compliance of the lung and decreased elastic recoil of the lung. As a result of the decreased elastic recoil, the equal pressure point forms early during expiration, resulting in trapping of air, and ultimately causing increased total lung capacity, functional residual capacity, and residual volume. In pulmonary function tests, expiratory flow rate and FEV_1 are reduced as a result.
10. **A.** The presence of surfactant at the air-fluid interface of alveoli and small airways results in lower surface tension and therefore increased pulmonary compliance, reducing the work of breathing. Surfactant contains the phospholipid dipalmitoyl phosphatidyl choline. Surfactant deficiency is responsible for respiratory distress syndrome of the newborn.
11. **A.** An increase in hematocrit will result in a proportional rise in the amount of oxygen bound to hemoglobin in blood. At 100 mm Hg PO_2 , hemoglobin is saturated with oxygen, and an increase in PO_2 will only result in a minor rise in oxygen content by raising the small amount of dissolved oxygen. Likewise, because hemoglobin in arterial blood is normally nearly saturated with oxygen, increased alveolar ventilation will have very little effect on oxygen content. An increase in 2,3-DPG or a fall in blood pH will shift the oxyhemoglobin dissociation curve to the right, resulting in a fall in bound oxygen.
12. **C.** The pH is below the normal level of 7.4, indicating acidosis; because PCO_2 is elevated, this is a case of respiratory acidosis (high PCO_2 is the cause of the low pH).
13. **D.** In acute adaptation to high altitude, hypoxemia stimulates respiratory rate. Heart rate is also elevated. 2,3-DPG is elevated in blood, resulting in right-shift of the oxyhemoglobin dissociation curve, causing oxygen to more readily dissociate from hemoglobin at the tissue level. Renal compensation will result in elevated plasma bicarbonate level. In the long term, however, increased hematocrit (higher red blood count [RBC] and hemoglobin concentration in blood) is an important compensatory mechanism, resulting in increased oxygen carrying capacity of blood.
14. **B.** Central chemoreceptors respond mainly to changes in arterial PCO_2 , which diffuses readily into the cerebrospinal fluid (CSF) and alters CSF pH, resulting in stimulation of respiration when arterial PCO_2 is elevated. The blood-brain barrier is largely impermeable to HCO_3^- or H^+ . Peripheral chemoreceptors respond to changes in arterial PO_2 and also pH and PCO_2 .
15. **D.** The initial, rapid adjustment of respiration during exercise is caused by input from proprioceptive afferents from joint receptors to the respiratory center in the brain, collaterals to the respiratory center from motor pathways for muscle activation, as well as additional, undefined factors. The additional elevation of respiration during continuing exercise is caused by feedback systems involving chemoreceptors and changes in body temperature.

Section 5: Renal Physiology

1. **C.** The clearance of inulin (C_{in}) is equated with the glomerular filtration rate, because inulin is freely filtered, is not reabsorbed or

secreted, and all filtered inulin is excreted. Thus, if the clearance of a freely filtered substance is less than C_{in} , it means that overall there was reabsorption (however, it does not determine whether secretion might also have occurred).

2. C. Increased extracellular matrix proteins thicken the glomerular basement membrane, increasing the filtration barrier, thus reducing the permeability to plasma.
3. A. Aldosterone is produced in the zona glomerulosa of the adrenal cortex.
4. B. Filtration fraction is defined as the glomerular filtration rate (GFR) divided by the renal plasma flow (RPF). The renal plasma flow equals $[RBF \times (1 - \text{hematocrit})]$, or 600 mL/min. Because C_{in} is equated with the GFR, the filtration fraction is 125 mL/min $\div$ 600 mL/min, or $\sim 20\%$.
5. A. Glucose is 100% reabsorbed in the proximal convoluted tubule, via Na^+ -glucose cotransporters.
6. D. Plasma antidiuretic hormone has no direct effect on renal potassium handling, while the other conditions cause either secretion (high dietary or plasma potassium and aldosterone) or enhanced reabsorption (low dietary or plasma potassium and acidosis).
7. D. Loop diuretics target the Na^+ - K^+ - 2Cl^- cotransporters on the thick ascending limb of Henle. When the transporters are blocked, the solutes are carried distally (where there is little sodium reabsorption in the absence of aldosterone) and most of the fluid is excreted as urine. Because this transporter is a key factor in the countercurrent multiplier system, an additional result of the use of loop diuretics is the washing out of the medullary interstitial concentration gradient, which contributes to the sustained diuresis.
8. E. In the collecting duct cells, binding of ADH to V2 receptors stimulates the insertion of aquaporins into the apical membranes. This causes sodium-free water reabsorption.
9. A. Distal sodium reabsorption has no effect on the medullary concentration gradient, whereas the other factors all play significant roles in creating and maintaining the gradient.
10. D. Free water clearance (C_{H_2O}) = $V - [(U_{osm}/P_{osm}) \times V]$, or +1. The positive value implies that water was cleared in excess of the amount required for iso-osmotic excretion of solutes present in the urine.
11. C. Angiotensin II has two direct actions on the kidneys, to increase proximal tubular sodium reabsorption and to constrict renal afferent and efferent arterioles. These actions increase sodium and water reabsorption. Renin is secreted from the juxtaglomerular cells in response to low sodium concentration and low tubular fluid flow rate in the distal tubule. Aldosterone stimulates sodium reabsorption in the late distal tubules and collecting ducts.
12. A. The reduction in vascular volume will stimulate sympathetic vasoconstriction and elevate sodium and fluid-retaining systems. Atrial natriuretic peptide is released from cardiac myocytes in response to increased atrial stretch during volume expansion. Thus, during dehydration, circulating ANP will be low.
13. E. Diabetes insipidus (DI) is usually of central origin (nephrogenic DI is rare), following trauma, disease, or surgery affecting the pituitary gland. Central DI involves the loss of ADH, so water channels are not present in the apical membranes of the collecting ducts, and urine cannot be concentrated. This leads to massive excretion (3 to 18 L/day) of hypotonic urine.
14. C. Plasma bicarbonate is low in metabolic acidosis, and the α -intercalated cells of the collecting ducts will increase H^+ secretion.
15. C. NAE is determined by the sum of urinary ammonium and titratable acids, minus any excreted bicarbonate. It does not depend on sodium excretion.
16. B. The pH is <7.4 , and therefore the condition is acidosis. Because the low plasma bicarbonate (<24 mEq/L) also reflects acidosis, the disorder is metabolic in origin. The anion gap can be used to determine whether the acidosis is from acid loading or base loss; $\text{AG} =$

$\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$, or $136 - 114 = 22$. The usual anion gap is ~ 8 to 12, and an increase in the gap reflects the addition of acid. Base loss (as found with diarrhea) would show no change in AG, because the loss of HCO_3^- in the stool would be matched by an increase in plasma Cl^- .

17. D. In respiratory alkalosis, the high pH will be matched by low PCO_2 (indicating a reduction in plasma "acid" as the primary disturbance). In *uncompensated* respiratory alkalosis, the plasma HCO_3^- will be normal.
18. A. Phosphoric acid (H_2PO_4^-) is the primary TA excreted in the urine. Phosphates make very good buffers in urine because their pK is near the urine pH and there is a large amount ($\sim 25\%$ of the filtered phosphate load) of dibasic phosphate (HPO_4^{2-}) available for conversion to TA and excretion.

Section 6: Gastrointestinal Physiology

1. D. The GI tract does not directly regulate systemic blood flow.
2. E. Growth hormone has no apparent effect on motility or secretion in the GI tract.
3. D. The GI tract has an intrinsic nervous system (enteric nerves), which responds to signals from luminal receptors and hormones, as well as extrinsic nerves. When the extrinsic nerve input is severed, motility and secretion continues, but not as efficiently.
4. B. Vagal afferent nerves regulate all of the actions EXCEPT primary esophageal peristalsis, which is mediated by the swallowing center in the medulla. The vagus does participate in secondary esophageal peristalsis, in conjunction with local enteric nerves.
5. B. Action (or spike) potentials in the GI tract are caused by the influx of calcium into the smooth muscle when the slow waves are depolarized above -40 mV. This mechanism is active throughout the tract, and depolarization can result from local stretch (mechanoreceptors acting on enteric nerves), extrinsic nerves, and peptides.
6. C. The undulations are a result of small changes in membrane potential due to the activity of the basolateral Na^+/K^+ ATPase. The slow waves occur at different rates from the mid-stomach through the colon.
7. D. Peristalsis follows the law of the intestines, where muscle is contracted proximal to the bolus and relaxed distal to the bolus, producing aboral movement of the chyme.
8. B. There is no evidence for involvement of the parasympathetic nervous system in the MMC.
9. D. Gastric inhibitory peptide (GIP) is an endocrine hormone secreted from cells in the duodenum and jejunum of the small intestine. The other substances are secreted directly into the gastric lumen.
10. C. Vagal afferent nerves stimulate gastric acid secretion directly at the parietal cell, and indirectly by increasing gastrin-releasing peptide (GRP) (and thus, gastrin release) and inhibiting somatostatin release.
11. E. Proton pump activity drives acid secretion and is the target for regulatory hormones, peptides, and nerves.
12. B. Sodium enters the luminal membrane by all of the mechanisms except the Na^+ pump (Na^+/K^+ ATPase), which is located on the basolateral membrane, and maintains the low intracellular Na^+ concentrations that drive the luminal Na^+ transport mechanisms.
13. C. Secretin is a hormone released from S-cells in the duodenum and early jejunum in response to acidic chyme. The secretin binds to receptors on the pancreatic acinar cells and stimulates release of electrolyte buffers into the pancreatic duct.
14. B. The liver performs all of the functions noted, except vitamin production. It does store a variety of important vitamins and minerals, including vitamin B12 and iron.
15. D. Obstruction of blood flow through the liver increases the pressure in the portal vein, which brings blood from the intestines to

the liver. This eventually results in portal hypertension. The obstruction also reduces bile secretion.

16. **A.** The hepatocytes are the primary site of formation of lipoproteins, which carry fats, protein, and cholesterol to different tissues for processing and storage.
17. **B.** Bile is amphipathic and essential for carrying the hydrophobic lipids through the unstirred water layer adjacent to the enterocytes. The ability to form micelles, with hydrophilic ends of bile oriented outward, and lipophilic ends oriented inward and associated with the lipids, allows the efficient transport of lipids to the intestinal cells.
18. **D.** Efficient digestion of proteins in the small intestine depends on the pancreatic proteases, which have optimal catalytic activity near physiologic pH (between pH 7 to 8). Low luminal pH will significantly decrease catalytic activity and reduce the amount of proteins digested to constituents that can be absorbed.
19. **D.** Starch is the primary dietary carbohydrate, and digestion begins in the mouth with salivary α -amylase. Only around 25% of digestion occurs preduodenally: pancreatic α -amylase digests the remaining starch to malto-oligosaccharides, and then intestinal brush border saccharidases (maltase, isomaltase, sucrose, and lactase) digest oligosaccharides and disaccharides (sucrose and lactose) to monosaccharides (glucose, galactose, and fructose).
20. **A.** The presence of chyme in the antrum of the stomach and the duodenum stimulates release of gastrin, which is a potent stimulator of gastric acid secretion. It does not promote buffer secretion.
21. **C.** Celiac sprue is an autoimmune reaction initiated by gluten proteins in wheat, resulting in the flattening of the villus lining and expansion of the crypts. The reduction in the upper villi severely decreases the brush border enzyme digestion of carbohydrates and proteins and reduces the surface area for absorption.
22. **B.** B12 is an essential nutrient and has multiple mechanisms protecting it from protease activity and facilitating entry into the intestinal cells. There is no limitation to absorption posed by intracellular B12 concentrations.

Section 7: Endocrine Physiology

1. **E.** Thyroid hormone, vitamin D, and steroid hormones are lipophilic and readily diffuse into target cells, where they are bound by nuclear receptors, initiating gene transcription. Peptide hormones and catecholamines are bound by membrane receptors, initiating an intracellular signaling cascade that ultimately leads to regulation of cellular function.
2. **E.** The cellular action of hormones is dependent on the concentration of hormone in the blood, as well as the number and affinity of hormone receptors in the cell or on the cell membrane. In addition, some hormones (particularly lipophilic hormones such as steroids) are carried in the blood by binding proteins, and it is specifically the free fraction of hormones in the blood that is available for binding by cellular receptors.
3. **B.** Vasopressin promotes increased water absorption by the kidney, and its release is stimulated by decreased blood volume, increased body fluid osmolality, and decreased blood pressure.
4. **A.** Oxytocin, a 9 amino acid peptide, is released directly into the general circulation by the posterior pituitary gland but is synthesized in the paraventricular and supraoptic nuclei of the hypothalamus. It is carried by axonal transport to nerve endings in the posterior pituitary. Its actions include milk expulsion during breastfeeding and uterine contraction, although a physiological role in induction or progression of labor is not well-established.
5. **E.** Growth hormone is released by the anterior pituitary somatotrophs, with several pulses occurring during sleep, as well as a basal level of secretion. Its release is stimulated by hypothalamic growth hormone-releasing hormone and inhibited by somatostatin. At the tissues, it acts by stimulating the synthesis of insulin-like growth factor.
6. **E.** Pituitary thyroid-stimulating hormone (TSH) is a primary stimulus for thyroid hormone (TH) release.
7. **E.** Hypothyroidism in adults results in myxedema, which is associated with nonpitting edema, hypotension, fat deposition, cold intolerance, and depression, among other symptoms.
8. **D.** Both T_3 and T_4 are released from the thyroid gland, but 20 times the amount of T_4 is released compared to T_3 . Most of the T_3 and T_4 is bound to proteins including albumin and thyroxine-binding globulin.
9. **C.** Feedback control of TH production is illustrated by elevated circulating free T_3 and T_4 , which reduces both pituitary thyroid-stimulating hormone (TSH) and hypothalamic thyrotropin-releasing hormone (TRH), resulting in reduced TH production.
10. **C.** Cortisol synthesis and release by the adrenal cortex is regulated by the anterior pituitary hormone ACTH. Although ACTH is required for synthesis of all adrenal steroids, androgen and aldosterone synthesis and release are primarily controlled by other mechanisms. Release of catecholamines by the adrenal medulla is controlled by the sympathetic nervous system.
11. **C.** The zona fasciculata secretes cortisol, which has anti-inflammatory and anti-immune effects among its many actions. The zona reticularis secretes androgens and the zona glomerulosa secretes aldosterone. The adrenal medulla secretes catecholamines.
12. **E.** Dehydroepiandrosterone and androstenedione are the major adrenal androgens. In females, for whom the adrenal gland is the only significant source of circulating androgens, all of the listed effects are stimulated by these hormones.
13. **C.** The primary stimuli for aldosterone secretion by the adrenal zona glomerulosa are hyperkalemia and angiotensin II. Aldosterone synthesis and secretion is inhibited by atrial natriuretic peptide.
14. **E.** The GLUT4 transporters are the insulin-sensitive transporters present on most cell membranes, and insertion of these transporters facilitates glucose entry.
15. **C.** GLUT2 transporters are prominent on the pancreas, small intestine, brain, and liver and facilitate entry of glucose into the cells. In the β -cell of the pancreas, glucose metabolism increases ATP and reduces K^+ efflux. This depolarizes the β -cell, causing Ca^{2+} influx and release of the insulin (and C-peptide).
16. **E.** All of the factors listed will stimulate insulin secretion.
17. **E.** Insulin has a hypoglycemic effect on the plasma by increasing GLUT4 transporters, which facilitate glucose entry into adipose tissue and cardiac and skeletal muscle, while it decreases gluconeogenesis and glycogenolysis and increases glycogen storage.
18. **D.** Decreased plasma calcium concentrations will stimulate the release of parathyroid hormone (PTH), which acts at bone to increase resorption, releasing calcium into the blood. PTH also acts at the kidney to increase calcium reabsorption.
19. **B.** PTH is released into plasma in response to reduced plasma calcium levels. At the kidneys it increases distal calcium reabsorption and reduces phosphate reabsorption. It also stimulates the renal conversion of 25-hydroxycholecalciferol to active vitamin D (1,25-dihydroxycholecalciferol), which increases intestinal absorption of calcium.
20. **E.** Primary hyperparathyroidism typically results from parathyroid tumors that have no negative feedback mechanisms and ultimately increases plasma calcium concentrations. The elevated PTH increases the renal conversion of 25-hydroxyvitamin D to active vitamin D, increasing intestinal calcium absorption; the reduction in fecal calcium can cause constipation. In addition, the PTH increases bone resorption (can cause bone pain), and the elevated filtered load of calcium to the kidney can cause stone formation (calculi). Overall, these symptoms can cause fatigue and general

malaise. These “stones, bones, moans, and groans” can be characteristic symptoms of hyperparathyroidism.

21. **B.** Differentiation of the external genitalia produces female genitalia in the absence of androgens. In the male fetus, the testes produce testosterone, which is converted to dihydrotestosterone (DHT) in the primitive genital structures. DHT stimulates formation of male genitalia.
22. **C.** The linear growth spurt that occurs during puberty is mainly attributed to estradiol in both boys and girls. Estrogenic hormones increase linear growth of long bones and also promote the eventual closure of the epiphyseal plates.
23. **A.** In developing follicles, androgen is synthesized and secreted by theca interna cells. The conversion of androgens to estradiol by the follicular granulosa cells is stimulated by luteinizing hormone (LH).
24. **B.** During fetal development, germ cells in the ovaries give rise to oogonia, which proliferate and begin meiosis. They become primary oocytes and are arrested at this stage.
25. **B.** FSH acts on Sertoli cells of the seminiferous tubules, stimulating androgen-binding protein synthesis. This protein binds testosterone, which promotes spermatogenesis. The Sertoli cells also synthesize inhibin, which inhibits follicle-stimulating hormone (FSH) secretion by the anterior pituitary gland.

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